

Price wedge anomaly signals: construction log

September 12, 2026

How to read this

One entry per characteristic. *Agreement* is the mean correlation, across the ten deciles and 462 formation months, between our decile capitalisation weights and the replication package's own (`datas_all_charP`). It tests the characteristic and the sort alone, independently of the 241-month tracking and the discount factor. Above 0.85 the construction is treated as verified. *Error* is our price wedge less the paper's Table 1 value, in percentage points.

Rejected alternatives records what was tried and did not work, so the same ground is not covered twice.

Summary

Signal	Group	Agreement	Long err.	Short err.	Status
OA	intangibles	0.963	-1.53	0.15	verified
AOA	intangibles	0.971	-1.15	0.05	verified
ATO	profitability	0.974	-0.86	0.16	verified
RNA	profitability	0.978	0.60	-0.30	verified
dSOUT	investment	0.992	-0.53	-0.68	verified
dPIA	investment	0.943	0.23	-0.87	verified
aSIZE	trading frictions	0.856	-14.69	0.21	verified
A2ME	value	0.972	3.57	-0.13	verified
AT	trading frictions	0.988	-0.02	0.11	verified
BEME	value	0.977	1.24	-0.70	verified
C2A	value	0.963	-0.06	-0.20	verified
C2D	value	0.959	-0.48	-2.03	verified
CAT	profitability	0.977	1.23	-0.32	verified
D2P	value	0.964	0.47	-0.36	verified
dCEQ	investment	0.951	-2.79	-1.51	verified
dGS	profitability	0.966	-0.69	-0.25	verified
dSO	value	0.966	-0.40	-0.53	verified
E2P	value	0.978	3.00	-0.41	verified
EPS	value	0.977	0.82	-3.88	verified
IDIOV	trading frictions	0.955	-0.21	-0.99	verified
I2A	investment	0.971	0.01	-1.30	verified
IPM	profitability	0.962	-0.78	-1.43	verified

Signal	Group	Agreement	Long err.	Short err.	Status
IVC	investment	0.970	-0.98	0.26	verified
SIZE	trading frictions	0.997	0.09	-0.21	verified
NOA	investment	0.967	-1.12	-0.26	verified
OL	intangibles	0.959	1.35	-0.16	verified
BETA_d	trading frictions	0.952	0.05	-0.17	verified
PCM	profitability	0.977	-0.44	-1.57	verified
PM	profitability	0.973	-0.61	-1.53	verified
PROF	profitability	0.978	-0.53	-0.29	verified
Q	value	0.978	0.91	-0.21	verified
MAXRET	trading frictions	0.984	-0.22	-1.16	verified
ROA	profitability	0.965	-0.06	0.53	verified
ROC	profitability	0.972	-0.95	0.24	verified
ROE	profitability	0.972	-0.41	-2.69	verified
ROIC	profitability	0.964	0.07	-0.23	verified
R_12_2	past returns	0.979	0.17	0.29	verified
R_12_7	past returns	0.984	-0.13	-0.43	verified
R_6_2	past returns	0.978	-0.02	-0.91	verified
R_2_1	past returns	0.986	-0.35	-0.45	verified
R_36_13	past returns	0.988	-0.95	-0.40	verified
S2C	value	0.968	0.15	-0.92	verified
S2P	value	0.979	5.01	-0.62	verified
SG	value	0.963	0.12	-0.82	verified
SAT	profitability	0.971	1.65	-0.20	verified
SPREAD	trading frictions	0.964	-1.48	2.38	verified
sdTURN	trading frictions	0.991	0.03	-0.06	verified
sdDVOL	trading frictions	0.896	-5.28	-0.08	verified
TAN	intangibles	0.978	-1.35	-1.38	verified
RETVOL	trading frictions	0.988	-0.14	-0.32	verified
DP	value	0.841	1.39	-3.39	unresolved
TNOVR	trading frictions	0.790	0.25	1.68	unresolved
SUV	trading frictions	0.073	3.76	-3.27	unresolved
DTO	trading frictions	0.363	1.37	-1.79	unresolved
aBEME	value	0.510	-0.81	-7.58	unresolved
aSAT	profitability	0.447	-3.51	3.04	unresolved
aPM	profitability	0.284	1.62	1.08	unresolved

Where the paper and its data disagree

Graded deliberately. Only category A claims something is wrong. B and C are documentation that cannot be built from; D concerns the reported results rather than the signals; E records things that look like discrepancies and are not, so they are not re-opened.

A. An error in the signal itself

The construction contradicts the formula the paper states, the wording of its own appendix, and the paper it cites, and the difference changes what the signal measures.

OA, AOA

Documented: Appendix Table A.1: "Operating Accruals: Changes in non-cash working capital minus depreciation (dp), all scaled by lagged total assets." Freyberger, Neuhierl and Weber word it identically and cite Sloan (1996), whose equation 1 subtracts depreciation.

Implemented: Depreciation sits inside the current-liabilities bracket, so expanding gives "+ dp": accruals are computed net of everything except depreciation, which is added.

Evidence: Netting depreciation out agrees with the package's own published decile weights at 0.286; adding it gives 0.963. Decile one's share of market capitalisation comes out at 5.29% against a published 5.30%, and the long-leg wedge error falls from +39.9 percentage points to -1.5. AOA, the absolute value, goes from 0.041 to 0.971.

Why it matters: OA and AOA are not operating accruals. Depreciation is large and persistent for capital-intensive firms, so those firms are sorted toward the high-accrual leg by their depreciation rather than away from it. The sign of the depreciation term is the whole content of Sloan's measure.

Note: Chen and Zimmermann's PctAcc.do carries the identical bracket placement while their Accruals.do does not, which suggests both inherited the same transcription rather than arriving at it independently.

B. The appendix cannot be built from

The implementation is defensible, often the only thing possible, but somebody working from Table A.1 alone would get a materially different series. Agreement scores below are what the literal reading achieves against what the data does.

SPREAD

Documented: "Average daily bid-ask spread in the previous month", citing Chung and Zhang (2014), whose estimator is the quoted spread.

Implemented: The daily high-low range, $2(\text{high}-\text{low})/(\text{high}+\text{low})$, averaged over the month.

Evidence: literal reading 0.02, implemented 0.94

Why it matters: Not a defect in the data – CRSP carries quotes for 9% of daily rows in the 1960s and 53% in the 1980s, so a quoted spread cannot be computed over this sample at all – but the citation points at an estimator the sample cannot support, and the substitution is unstated.

sdDVOL

Documented: "Standard deviation of residuals from a regression of daily volume (vol) on a constant", i.e. the dispersion of daily share volume in levels.

Implemented: The standard deviation of $\log(1 + \text{daily} * \text{dollar} * \text{volume})$.

Evidence: share volume in levels 0.06, log share volume 0.63, log dollar volume 0.87

Why it matters: Two undocumented choices at once, the units and the logarithm, and in levels the measure captures firm size rather than variability.

dSOUT

Documented: "Annual % change in shares outstanding (shrout)."

Implemented: Growth in shrout multiplied by CRSP's cumulative share adjustment factor.

Evidence: literal reading 0.51, implemented 0.97

Why it matters: Taken literally the signal reads every two-for-one split as a 100% issuance. This is the largest gap between a literal reading and the data of any signal outside category A.

ROIC

Documented: "...to the sum of common equity (ceq), total liabilities (lt), and cash and short-term investments (che)."

Implemented: Cash is subtracted from invested capital, not added.

Evidence: as worded 0.64, as implemented 0.93

Why it matters: Here the wording is wrong and the implementation is right – invested capital is capital tied up in the business, net of cash – so this is a typo in the appendix rather than a problem with the data.

BETA_d

Documented: "The sum of the regression coefficients of daily excess returns on market excess returns and the lag of market excess returns." No estimation window is given.

Implemented: Estimated over twelve months of daily data.

Evidence: one month 0.35, six 0.62, twelve 0.94, twenty-four 0.63

Why it matters: The window is the difference between a usable signal and noise, and it is not stated. Separately, FNW's own entry under this name describes the Frazzini-Pedersen beta – a correlation times a volatility ratio – which is a different estimator from the Dimson sum A.1 describes.

DP

Documented: "Sum of monthly dividend over the last 12 months to last month's price (prc)."

Implemented: Dividends in dollars over the *current* month's market capitalisation.

Evidence: per share over price 0.84, dollars over market cap 0.92

Why it matters: Minor, but per share the share count drifts through the year and the two are not the same quantity.

TAN

Documented: The formula is given; the missing-data rule is not.

Implemented: Every component required non-missing.

Evidence: zero-filling 0.48, requiring all 0.96

Why it matters: Zero-filling reads as "this firm holds no tangible assets" and sends firms with incomplete filings to the bottom decile.

C. The appendix drops a qualifier its own source insists on

Table A.1 is a condensation of Freyberger, Neuhierl and Weber. In a few places the condensation loses something that changes the answer.

A2ME, E2P, S2P, BEME, SIZE

Documented: FNW state the market capitalisation timing explicitly – "as of December t-1" for A2ME, E2P and BEME, "as of December" for S2P, "of the previous month" for SIZE. Table A.1 says only "market capitalization (prc x shrou)" with no timing at all.

Implemented: The current month's market capitalisation.

Evidence: contemporaneous 0.95-0.99 across the five, December of the prior year 0.61-0.97

Why it matters: The data does not follow FNW here, and A.1 is silent rather than wrong. But anyone building these from FNW, as the citation invites, would use December and land a long way off – Q falls from 0.98 to 0.61, E2P from 0.97 to 0.69.

Q

Documented: FNW write "the market value of equity (SHROUT times PRC) minus cash and short-term investments (CEQ)" – CEQ is common equity, not cash.

Implemented: Common equity, which is what CEQ means.

Why it matters: A typo in FNW that Table A.1 silently corrects. Recorded because it runs the other way.

D. Issues in the reported results rather than the signals

Found while replicating the paper's own tables and out-of-sample tests.

Table of characteristics-to-wedge mappings

Documented: Profitability alone is reported with an R-squared of exactly 0.00, alongside a non-zero coefficient of +0.13.

Implemented: We obtain 0.18 on the same 114 extreme portfolios.

Evidence: every other entry in that table replicates within a few points: PC3 0.705 against 0.76, PC6 0.752 against 0.77, FF5 0.704 against 0.74, BEME 0.550 against 0.67

Why it matters: A zero R-squared with a non-zero slope is hard to produce and is probably a reporting slip.

aPM, dGS

Documented: Each is signed so its long-short earns a positive one-month alpha, and reported with a long and a short leg.

Implemented: Their one-month alphas in our data are 0.00012 and 0.0013 a month.

Evidence: our sign convention agrees with the package for 55 of 57 characteristics; these are the two that differ

Why it matters: For aPM especially the alpha is indistinguishable from zero, so which leg is "long" is arbitrary and the published long and short wedges for it could swap without anything being wrong.

The out-of-sample firm-level correlation of 0.98

Documented: Firm wedges from a pre-1998 mapping correlate 0.98 with full-sample firm wedges, offered as evidence the approach is robust out of sample.

Implemented: Both series are linear functions of the same three principal components of the same firm characteristics, differing only in coefficients.

Evidence: Refitting the mapping to portfolio wedges with HALF the spread gives firm wedges correlating 1.0000 with the correct ones; fitting to wedges plus noise of equal variance gives 0.9976.

Why it matters: A mapping that gets every magnitude wrong by a factor of two passes this test perfectly. It establishes that the coefficient vector is stable, which is a real if modest claim, but it never compares a prediction against a realised wedge.

The out-of-sample portfolio correlation of 0.92

Documented: First-half portfolio wedges correlate 0.92 with full-sample wedges.

Implemented: The first half is one half of the full sample, so the two overlap by construction.

Evidence: For two halves correlating rho, each correlates with the whole at about $\sqrt{((1+\rho)/2)}$. At the paper's footnoted rho of 0.77 that is 0.94; at ours, 0.58, it is 0.89 and we observe 0.95.

Why it matters: The 0.92 is close to implied by the 0.77 in the footnote. The footnote is the informative number, and it is the one we cannot reproduce.

The mapping R-squared of 0.76

Documented: Three principal components explain 0.76 of portfolio price wedges.

Implemented: Computed on the 114 extreme deciles only.

Evidence: on all 570 portfolios the same mapping gives 0.543 in sample and 0.411 leaving a characteristic family out; the direct rank regression gives 0.822 and 0.749 on the same portfolios

Why it matters: Restricting to the extremes flatters three principal components specifically – the gap to a direct regression is 0.09 on the extremes and 0.31 on all ten deciles. Most firms sit in the middle deciles.

Time-split out-of-sample tests

Documented: Wedges are recomputed on subsamples of formation dates.

Implemented: The market price of risk is calibrated once on the full sample and carried over.

Evidence: Carried over, early cohorts average -37 percentage points and late ones +33, a seventy-point flip that is purely the difference in realised market returns across the cut. Re-solving it within each half – 3.9 to 4.9 early against 2.4 to 2.9 late – the flip disappears and root mean squared error falls from 37-77 points to 10-15.

Why it matters: Correlations are location-invariant so the paper's reported correlations are unaffected. It does matter for the Panel B investment-q regressions, which use out-of-sample firm wedges as a regressor in levels.

E. Looks like a discrepancy, is not

Recorded so these are not re-opened. Each cost real time to resolve.

The five momentum sorts

Documented: "Cumulative return from 12 months to 2 months ago."

Implemented: The panel stores, on row t , the return over months $t-11$ to $t-1$ – one month later than the name. The sort then lags every characteristic by a further month, so the signal a portfolio formed at t actually sorts on is the return from $t-12$ to $t-2$, exactly as documented.

Evidence: storing the literal window instead drops agreement from 0.97 to 0.65, and for $R_{2.1}$ from 0.97 to 0.11

Why it matters: NOT an error. The economics is right and the stated definition is what gets used; only the storage convention is undocumented. It matters enormously for anyone rebuilding the panel and not at all for the paper's results.

R_12_2 stored as a gross return

Documented: A cumulative return.

Implemented: Stored as $1 + r$ rather than r ; the median difference against ours is exactly 1.0000.

Why it matters: A units convention. Ranks are untouched, so no sort is affected.

Market capitalisation units

Implemented: Stored in millions in the package and in thousands in CRSP, a factor of exactly 1000.

Why it matters: Value weighting is scale-invariant, so nothing is affected.

The panel's row alignment

Implemented: MainPart1.m slices the panel from one row *before* July 1926, so its rows are offset by a month relative to the obvious reading.

Evidence: Comparing without noticing makes contemporaneous market equity look like a 3% match and lagged market equity a 91% match, which is backwards. Implementing the lagged version accordingly made the sorts worse – 50 verified characteristics down to 48, BEME from 0.97 to 0.84 – which is what exposed the error.

Why it matters: A trap for anyone comparing against the shipped panel, not a problem with the paper.

Construction, signal by signal

OA — Operating accruals

Group intangibles **Source** Sloan (1996), per Appendix Table A.1 **Frequency** annual **Agreement** 0.963

Inputs. act, che, lct, dlc, txp, dp, at

Formula.

$$OA = (d\ act - d\ che - (d\ lct - d\ dlc - d\ txp - dp)) / at[t-1]$$

where $d\ txp$ treats a missing value as zero.

Note the placement of dp : it sits INSIDE the bracket, so expanding gives

$OA = d\ act - d\ che - d\ lct + d\ dlc + d\ txp + dp$, i.e. depreciation is ADDED to accruals.

Notes. This is not Sloan (1996) and not what Table A.1’s own wording says (“changes in non-cash working capital minus depreciation”). It is nevertheless unambiguously what the paper’s data does: netting depreciation out gives agreement 0.286, adding it gives 0.963, and decile one’s share of market capitalisation comes out at 5.29% against a published 5.30%. Chen and Zimmermann’s PctAcc.do carries the identical bracket placement while their Accruals.do does not, which suggests a shared transcription. WORTH RAISING WITH THE COAUTHORS: as published, OA and AOA are not accruals net of depreciation, so capital-intensive firms sort toward the high-accrual leg because of their depreciation rather than away from it.

Rejected alternatives.

Variant	Agreement	Verdict
Sloan’s grouping with dp subtracted outside the bracket	0.286	rejected
literal reading of A.1, all three current-liability terms subtracted	0.065	rejected
dp subtracted, scaled by average rather than lagged total assets	0.251	rejected
dp subtracted, every delta zero-filled	0.163	rejected
change in net operating assets	0.234	rejected
change in working capital from $wcap$ directly	-0.058	rejected
cash-flow statement form, ni less $oancf$, over average assets	0.087	rejected
cash-flow form with balance-sheet fallback	0.269	rejected
scaled by sales, by absolute earnings, by current assets	0.154	rejected
winsorised 1/99 monthly, or —OA— capped at 1, 0.5, 0.3	0.065	no effect; outliers are not the cause
JKP $oaccruals_at$ taken directly from their library	0.278	better than ours was, worse than the fix
shifted +/- 6 months against the published series	–	no peak; not a timing offset

AOA — Absolute operating accruals

Group intangibles **Source** Bandyopadhyay, Huang and Wirjanto (2010) **Frequency** annual
Agreement 0.971

Inputs. act , che , lct , dlc , txp , dp , at

Formula.

$AOA = \text{—}OA\text{—}$, with OA exactly as above (depreciation added, not subtracted).

Notes. Follows OA mechanically. Neither Chen and Zimmermann nor Jensen, Kelly and Pedersen carry a Bandyopadhyay signal, so there is no third implementation to compare against; it is verified only through OA .

Rejected alternatives.

Variant	Agreement	Verdict
absolute value of every rejected OA variant above	0.087	rejected

ATO — Asset turnover

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.974

Inputs. sale, at, che, ivao, dlc, dlтт, mib, pstk, ceq

Formula.

operating assets = at - che - ivao (che, ivao missing -і 0)
operating liabilities = at - dlc - dlтт - mib - pstk - ceq
(dlc, dlтт, mib, pstk missing -і 0; ceq REQUIRED)
net operating assets = operating assets - operating liabilities
ATO = sale / NOA[t-1]

Notes. Common equity must be required rather than zero-filled. Filled, operating liabilities collapse to assets less debt, net operating assets go to nearly zero, and anything divided by them explodes into the end deciles. Agreement 0.78 filled against 0.92 required. This is Chen and Zimmermann's convention in RetNOA.do, where dlc, dlтт, mib and pstk are zero-filled and ceq is not.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled like the other liability terms	0.780	rejected
contemporaneous rather than lagged net operating assets	0.582	rejected
operating assets without subtracting ivao	0.558	rejected
observations with non-positive NOA discarded	0.494	rejected
Soliman's component-built operating assets (rect+invт+aco+ppent+intan-ap-lco-lo), averaged	0.241	rejected
lagged sales over lagged NOA	0.625	rejected

RNA — Return on net operating assets

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.978

Inputs. oiadp, at, che, ivao, dlc, dlтт, mib, pstk, ceq

Formula.

RNA = oiadp / NOA[t-1], with NOA exactly as defined under ATO
(ceq required, the other liability terms zero-filled).

Notes. Same common-equity point as ATO: 0.76 with ceq filled, 0.95 required.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled	0.758	rejected
contemporaneous NOA	0.690	rejected
operating assets without ivao	0.551	rejected
non-positive NOA discarded	0.538	rejected
Soliman component-built operating assets	0.411	rejected

dSOUT — Change in shares outstanding

Group investment **Source** Pontiff and Woodgate (2008) **Frequency** monthly **Agreement** 0.992

Inputs. shrout, mthcumfacshr

Formula.

$$\begin{aligned}\text{adjusted shares } s[t] &= \text{shrout}[t] * \text{mthcumfacshr}[t] \\ \text{dSOUT}[t] &= 100 * (s[t] / s[t-12] - 1)\end{aligned}$$

Notes. CRSP's share counts are not split-adjusted, so raw growth reads every split as a 100% issuance. The cumulative share factor is the exact adjustment.

Rejected alternatives.

Variant	Agreement	Verdict
raw shrout growth, unadjusted for splits	0.513	rejected
split adjustment backed out of market cap and the ex-dividend return	0.883	close, superseded

dPIA — Change in plant and inventory over assets

Group investment **Source** Lyandres, Sun and Zhang (2008) **Frequency** annual **Agreement** 0.943

Inputs. ppgt, invt, at

Formula.

$$\begin{aligned}\text{dPIA} &= (\Delta \text{ ppgt} + \Delta \text{ invt}) / \text{at}[t-1], \\ &\text{where each delta treats a missing component as no change (zero) rather than} \\ &\text{discarding the firm.}\end{aligned}$$

Rejected alternatives.

Variant	Agreement	Verdict
both deltas required non-missing	0.777	rejected
scaled by average rather than lagged assets	0.771	rejected
net rather than gross plant (ppent)	0.493	rejected
scaled by lagged (ppgt + invt)	0.420	rejected

aSIZE — Industry-adjusted size

Group trading frictions **Source** Asness, Porter and Stevens (2000) **Frequency** monthly **Agreement** 0.856

Inputs. cap, siccd

Formula.

aSIZE = SIZE - mean(SIZE) within the Fama-French 48 industry that month,
equal-weighted over ordinary common shares.

Notes. Just above the threshold at 0.85, but its wedge is still 13 points from the paper's on the long leg – the widest gap of any characteristic that passes the construction check. Treat as unresolved in practice.

Rejected alternatives.

Variant	Agreement	Verdict
Fama-French 10 and 30 industries	0.866	marginally better than 48, within noise
industry median instead of mean	0.877	marginally better, not adopted
log size demeaned within industry	0.754	rejected
partial adjustment at 0.75	0.872	marginally better, not adopted
capitalisation-weighted industry mean	0.317	rejected

A2ME — Total assets (at) over market capitalization (prc x shrout)

Group value **Source** bhandari1988 **Frequency** monthly **Agreement** 0.972

Inputs. at

Formula.

at / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.700	rejected

AT — Total assets (at)

Group trading frictions **Source** gandhi2015 **Frequency** annual **Agreement** 0.988

Inputs. at

Formula.

Total assets.

BEME — Book equity to market equity. Book equity is shareholders' equity (seq

Group value **Source** – **Frequency** monthly **Agreement** 0.977

Inputs. at, ceq, lt, prc, pstk, pstkl, pstrkrv, seq, shrout, txditc

Formula.

be / me, with negative book equity discarded before sorting.

Book equity = seq, else ceq+pstk, else at-lt; plus txditc; minus the first available of pstrkrv, pstkl, pstk. Market equity is CRSP mthcap for the CURRENT month, in millions.

Notes. FNW state market equity as of December t-1 explicitly. The package's own stored panel says otherwise once its one-month row offset is accounted for: lined up correctly the current month matches 94.9% of firm-months exactly and December 3.9%. Implementing December drops agreement from 0.97 to about 0.70.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December of the prior year	0.700	rejected
market equity lagged one month	0.840	rejected; an artifact of a row-offset error in the comparison
market equity summed across share classes of the same permco	–	marginally better on values (95.7% vs 94.9%), not adopted

C2A — Cash and short-term investments (che) to total assets (at)

Group value **Source** palazzo2012 **Frequency** annual **Agreement** 0.963

Inputs. at, che

Formula.

che / at.

C2D — Cashflow to debt. Cashflow is the sum of income and extraordinary item

Group value **Source** ou1989 **Frequency** annual **Agreement** 0.959

Inputs. dp, ib, lt

Formula.

$$(ib + dp) / lt.$$

CAT — Sales (sale) to lagged total assets (at)

Group profitability **Source** haugen1996 **Frequency** annual **Agreement** 0.977

Inputs. at, sale

Formula.

$$sale / at[t-1].$$

D2P — Debt to price. Debt is long-term debt (dltt) plus debt in current liab

Group value **Source** litzenberger1979 **Frequency** monthly **Agreement** 0.964

Inputs. dlc, dltt, prc, shrout

Formula.

$$(dltt + dlc) / me, \text{ each debt term missing treated as zero.}$$

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.710	rejected

dCEQ — Annual % change in book value of equity (ceq)

Group investment **Source** richardson2005 **Frequency** annual **Agreement** 0.951

Inputs. ceq

Formula.

$$\text{Percent change in ceq.}$$

dGS — % change in gross margin minus % change in sales (sale). Gross margin

Group profitability **Source** abarbanell1997 **Frequency** annual **Agreement** 0.966

Inputs. cogs, sale

Formula.

$$\text{Percent growth in gross margin (sale-cogs) less percent growth in sale.}$$

dSO — Log change in the product of shares outstanding (csho) and the adjustm

Group value **Source** fama2008 **Frequency** annual **Agreement** 0.966

Inputs. ajax, csho

Formula.

$\log(\text{csho} * \text{ajax} / (\text{csho} * \text{ajax})_{[t-1]})$, i.e. growth in split-adjusted Compustat share count.

E2P — Income before extraordinary items (ib) to market capitalization (prc x

Group value **Source** basu1983 **Frequency** monthly **Agreement** 0.978

Inputs. ib

Formula.

ib / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December t-1	0.690	rejected

EPS — Income before extraordinary items (ib) to shares outstanding (shrout)

Group value **Source** basu1977 **Frequency** monthly **Agreement** 0.977

Inputs. ib, shrout

Formula.

ib / (shrout/1000), i.e. earnings per share on CRSP's share count.

IDIOV — Standard deviation of the residuals from a regression of excess return

Group trading frictions **Source** ang2006 **Frequency** monthly **Agreement** 0.955

Formula.

Standard deviation of the residuals from a regression of daily excess returns on the three Fama-French factors, within the month, at least 15 days.

Notes. Emphatically monthly, unlike BETA_d: 0.93 at one month against 0.58 at twelve. JKP corroborate, carrying ivol_ff3_21d at a 21-day window, and our series correlates 0.997 with theirs.

Rejected alternatives.

Variant	Agreement	Verdict
estimated over twelve months	0.582	rejected

I2A — Annual % change in total assets (at)

Group investment **Source** – **Frequency** annual **Agreement** 0.971

Inputs. at

Formula.

Percent change in at.

IPM — Pre-tax income (pi) over sales (sale)

Group profitability **Source** – **Frequency** annual **Agreement** 0.962

Inputs. pi, sale

Formula.

pi / sale.

IVC — Annual change in inventories (invt) in the last two fiscal years over

Group investment **Source** thomas2002 **Frequency** annual **Agreement** 0.970

Inputs. at, invt

Formula.

$(\text{invt} - \text{invt}[t-1]) / ((\text{at} + \text{at}[t-1])/2).$

SIZE — Price (prc) times shares outstanding (shrout)

Group trading frictions **Source** fama1992 **Frequency** monthly **Agreement** 0.997

Inputs. prc, shrout

Formula.

Market equity, CRSP mthcap for the current month, in millions.

NOA — Operating assets minus operating liabilities to lagged total assets (a

Group investment **Source** hirshleifer2004 **Frequency** annual **Agreement** 0.967

Inputs. at, ceq, che, dlc, dltd, ivao, mib, pstk

Formula.

(operating assets - operating liabilities) / at[t-1], with operating assets and liabilities as defined under ATO.

Notes. Unaffected by the common-equity question that matters so much for ATO and RNA, because NOA uses the level rather than dividing by it.

Rejected alternatives.

Variant	Agreement	Verdict
ceq zero-filled	0.917	no effect

OL — Sum of cost of goods sold (cogs) and selling, general and administrati

Group intangibles **Source** novy2011 **Frequency** annual **Agreement** 0.959

Inputs. at, cogs, xsga

Formula.

(cogs + xsga) / at, each missing term treated as zero.

BETA_d — The sum of the regression coefficients of daily excess returns on mark

Group trading frictions **Source** lewellen2006 **Frequency** monthly **Agreement** 0.952

Formula.

Sum of the coefficients on the market and its lag, from a regression of daily excess returns on both, estimated over TWELVE months of daily data.

Notes. Twelve months, not one. A one-month Dimson beta is too noisy to sort on: 0.35 at one month, 0.62 at six, 0.94 at twelve, 0.63 at twenty-four. The moments are additive, so a twelve-month cross-product matrix is the sum of twelve monthly ones.

Rejected alternatives.

Variant	Agreement	Verdict
estimated over one month	0.353	rejected
six months	0.617	rejected
twenty-four and sixty months	0.631	rejected

PCM — Net sales (sale) minus cost of goods sold (cogs) all scaled by net sal

Group profitability **Source** gorodnichenko2016 **Frequency** annual **Agreement** 0.977

Inputs. cogs, sale

Formula.

$(\text{sale} - \text{cogs}) / \text{sale}.$

PM — Operating Income after depreciation (oiadp) to sales (sale)

Group profitability **Source** soliman2008 **Frequency** annual **Agreement** 0.973

Inputs. oiadp, sale

Formula.

$\text{oiadp} / \text{sale}.$

Rejected alternatives.

Variant	Agreement	Verdict
net income over revenue, as Chen and Zimmermann implement Soliman	0.387	rejected
oiadp over average sales	0.732	rejected

PROF — Gross profitability (gp) over book equity as defined in BEME

Group profitability **Source** ball2015 **Frequency** annual **Agreement** 0.978

Inputs. gp

Formula.

gp / be , with book equity as defined under BEME.

Q — Total assets (at) plus market value of equity (shrout x prc) minus boo

Group value **Source** – **Frequency** monthly **Agreement** 0.978

Inputs. at, ceq, txdb

Formula.

$(\text{at} + \text{me} - \text{ceq} - \text{txdb}) / \text{at}$, txdb missing treated as zero, me the current month's market equity in millions.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December of the prior year	0.610	rejected

MAXRET — Maximum daily return in the previous month

Group trading frictions Source bali2011 Frequency monthly Agreement 0.984
Formula.

Maximum daily return within the month.

Notes. Correlates 1.000 with JKP's rmax1_21d.

ROA — Income before extraordinary items (ib) to lagged total assets (at)

Group profitability Source balakrishnan2010 Frequency annual Agreement 0.965
Inputs. at, ib
Formula.

$ib / at[t-1]$.

ROC — Market value of equity (shrout x prc) plus long-term debt (dltt) minus

Group profitability Source chandrashekar2009 Frequency monthly Agreement 0.972
Inputs. at, che, dltt
Formula.

$(me + dltt - at) / che$.

ROE — Income before extraordinary items (ib) to lagged book-value of equity

Group profitability Source haugen1996 Frequency annual Agreement 0.972
Inputs. ib
Formula.

$ib / be[t-1]$.

ROIC — Earnings before interest and taxes (ebit) less non-operating income (n

Group profitability Source brown2007 Frequency annual Agreement 0.964
Inputs. ceq, che, ebit, lt, nopi
Formula.

$(ebit - nopi) / (ceq + lt - che)$, nopi missing treated as zero.

Notes. Appendix Table A.1 words the denominator as a sum, which would add cash to invested capital. Subtracting it is both what invested capital means and what the data says: 0.64 adding against 0.93 subtracting.

Rejected alternatives.

Variant	Agreement	Verdict
cash added to invested capital, as A.1 words it	0.640	rejected
every component required non-missing	0.640	no effect

R_12.2 — Cumulative return from 12 months to 2 months ago

Group past returns **Source** fama1996 **Frequency** monthly **Agreement** 0.979
Formula.

Compound return over months t-11..t-1, stored on row t.

Notes. Stored ONE MONTH EARLIER than the name suggests. The sort's own one-month lag then produces the stated twelve-to-two window. Storing the literal window drops agreement from 0.97 to 0.65.

Rejected alternatives.

Variant	Agreement	Verdict
literal window, months t-12..t-2	0.647	rejected

R_12.7 — Cumulative return from 12 months to 7 months ago

Group past returns **Source** novy2012 **Frequency** monthly **Agreement** 0.984
Formula.

Compound return over months t-11..t-6, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-12..t-7	0.533	rejected

R_6.2 — Cumulative return from 6 months to 2 months ago

Group past returns **Source** jegadeesh1993 **Frequency** monthly **Agreement** 0.978
Formula.

Compound return over months t-5..t-1, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-6..t-2	0.507	rejected

R.2.1 — Lagged one month return

Group past returns **Source** jegadeesh1990 **Frequency** monthly **Agreement** 0.986
Formula.

The CURRENT month's return, stored on row t.

Notes. The one-month shift matters most here: with a single month there is no overlap to rescue a timing error, and the literal 'lagged one-month return' gives 0.11 against 0.97.

Rejected alternatives.

Variant	Agreement	Verdict
lagged one-month return, as A.1 words it	0.108	rejected

R.36.13 — Cumulative return from 36 months to 13 months ago

Group past returns **Source** de1985 **Frequency** monthly **Agreement** 0.988
Formula.

Compound return over months t-35..t-12, stored on row t.

Rejected alternatives.

Variant	Agreement	Verdict
literal window t-36..t-13	0.744	rejected

S2C — Net sales (sale) to cash and short-term investments (che)

Group value **Source** ou1989 **Frequency** annual **Agreement** 0.968
Inputs. che, sale
Formula.

sale / che.

S2P — Net sales (sale) to market capitalization (shrout x prc)

Group value **Source** lewellen2015 **Frequency** monthly **Agreement** 0.979

Inputs. sale

Formula.

sale / me.

Rejected alternatives.

Variant	Agreement	Verdict
market equity as of December	0.720	rejected

SG — % growth rate in sales (sale)

Group value **Source** lakonishok1994 **Frequency** annual **Agreement** 0.963

Inputs. sale

Formula.

Percent change in sale.

SAT — Sales (sale) to total assets (at)

Group profitability **Source** soliman2008 **Frequency** annual **Agreement** 0.971

Inputs. at, sale

Formula.

sale / at.

Rejected alternatives.

Variant	Agreement	Verdict
sale over average assets	0.786	rejected
sale over lagged assets	0.736	rejected

SPREAD — Average daily bid-ask spread in the previous month

Group trading frictions **Source** chung2014 **Frequency** monthly **Agreement** 0.964

Formula.

Mean over the month of the daily high-low range, $2 \cdot (\text{high} - \text{low}) / (\text{high} + \text{low})$,
requiring at least 15 days.

Notes. NOT the quoted bid-ask spread, although FNW cite Chung and Zhang (2014). CRSP carries quotes for 9% of daily rows in the 1960s and 53% in the 1980s, so a quoted spread cannot be computed

over this sample: 0.02 against 0.94.

Rejected alternatives.

Variant	Agreement	Verdict
quoted bid-ask spread, $2 \times (\text{ask} - \text{bid}) / (\text{ask} + \text{bid})$	0.017	rejected
high-low range over the closing price	0.933	marginally worse

sdTURN — Standard deviation of residuals from a regression of daily turnover on

Group trading frictions **Source** chordia2001 **Frequency** monthly **Agreement** 0.991

Inputs. shrout, vol

Formula.

Standard deviation within the month of daily turnover (volume over shares outstanding), requiring at least 15 days.

Rejected alternatives.

Variant	Agreement	Verdict
coefficient of variation of turnover (Chordia et al.)	0.004	rejected

sdDVOL — Standard deviation of residuals from a regression of daily volume (vol

Group trading frictions **Source** chordia2001 **Frequency** monthly **Agreement** 0.896

Inputs. vol

Formula.

Standard deviation within the month of $\log(1 + \text{daily dollar volume})$, requiring at least 15 days.

Notes. In LOGS. In levels the dispersion measures scale rather than variability: 0.06 against 0.87.

Rejected alternatives.

Variant	Agreement	Verdict
standard deviation of dollar volume in levels	0.057	rejected
standard deviation of log share volume	0.632	rejected
coefficient of variation of dollar volume (Chordia et al.)	0.521	rejected

TAN — Tangibility is defined as $(0.715 \times \text{total receivables (rect)} + 0.547 \times$

Group intangibles **Source** hahn2009 **Frequency** annual **Agreement** 0.978

Inputs. at, che, invt, ppent, rect

Formula.

$$(0.715 \cdot \text{rect} + 0.547 \cdot \text{invt} + 0.535 \cdot \text{ppent} + \text{che}) / \text{at, every component REQUIRED.}$$

Notes. Zero-filling a missing component reads as 'this firm holds no tangible assets' and sends firms with incomplete filings to the bottom decile: 0.48 against 0.96.

Rejected alternatives.

Variant	Agreement	Verdict
missing components filled with zero	0.478	rejected

RETVOL — Standard deviation of residuals from a regression of excess returns on

Group trading frictions **Source** ang2006 **Frequency** monthly **Agreement** 0.988

Formula.

Standard deviation of daily excess returns within the month, at least 15 days.

Notes. Correlates 1.000 with JKP's rvol_21d.

DP — Dividend yield

Group value **Source** Litzenberger and Ramaswamy (1979) **Frequency** monthly **Agreement** 0.841

Inputs. ret, retx, cap

Formula.

dollars paid in month $s = (\text{ret}[s] - \text{retx}[s]) * \text{cap}[s-1]$
 $\text{DP}[t] = \text{sum of those dollars over months } t-11..t, \text{ divided by cap}[t],$
 requiring at least six of the twelve months.

Notes. Dividends in dollars over market capitalisation, not dividends per share over price. Per share the share count drifts through the year and agreement falls to 0.841. JKP's div12m_me, which is this same quantity, scores 0.928 against the package, and building it natively gives 0.924.

Rejected alternatives.

Variant	Agreement	Verdict
dividends per share over price (twelve-month sum)	0.841	rejected
per-share dividends restated for splits before summing	0.735	rejected; the more coherent calculation agrees worse
sum of twelve monthly dividend yields, no price	0.743	rejected

Variant	Agreement	Verdict
cash distributions from crsp.stkdistributions over price	0.348	rejected despite being the more principled source
dollar dividends over lagged rather than current market cap	0.869	rejected
JKP div_me (one month), div6m_me, div_at	0.652	rejected

TNOVR — Share turnover

Group trading frictions **Source** Datar, Naik and Radcliffe (1998) **Frequency** monthly **Agreement** 0.790

Inputs. vol, shrout

Formula.

$$\text{TNOVR}[t] = (\text{vol}[t] + \text{vol}[t-1] + \text{vol}[t-2]) / (3 * \text{shrout}[t])$$

Notes. Averaged over three months. Table A.1 reads as one month and the source paper does not say; three tracks the published weights best. Still below the verification threshold at 0.79, so this one is not settled.

Rejected alternatives.

Variant	Agreement	Verdict
single month, vol/shrout	0.707	rejected
twelve-month average	0.699	rejected
mean daily turnover within the month	0.706	rejected
Nasdaq volume scaled down 50% pre-1997 and 38% after	0.679	rejected
dollar volume over market cap	0.708	rejected
JKP turnover_126d	0.713	rejected

SUV — Standardised unexplained volume

Group trading frictions **Source** Garfinkel (2009) **Frequency** monthly (daily inputs) **Agreement** 0.073

Inputs. dlyvol, dlyret

Formula.

Within a month, regress $\log(1 + \text{daily volume})$ on a constant and the absolute values of positive and negative daily returns:
 $\ln(1+v_i) = a + b * \max(r_i, 0) + c * -\min(r_i, 0) + e_i$
Estimated on the previous month and applied to the current one:
 $\text{SUV}[t] = (\sum_i \ln(1+v_i) - \text{predicted}) / (n * \text{sd}(e))$
with predicted and sd(e) from month t-1's fit. At least 15 days required.

Notes. UNRESOLVED at 0.07. Neither Chen and Zimmermann nor JKP carry a Garfinkel signal, so

there is no reference implementation anywhere. The level of the decile weights comes out close (8.85 against a published 8.94 on decile one) while the month-to-month variation does not, which is the signature of a measure that is right in construction but wrong in timing or in which residual is taken. Full daily data has been pulled to test day-level variants the monthly moments cannot express.

Rejected alternatives.

Variant	Agreement	Verdict
regression in volume levels rather than logs	0.080	rejected
estimation window of 1, 6, 12 and 24 months applied out of sample	0.092	rejected at every length
previous month taken on a dense grid so the lag cannot cross a gap	0.072	rejected
in-sample residual dispersion rather than an out-of-sample residual	0.002	rejected
standardised by $\sqrt{n}$ rather than n	0.073	rejected

DTO — Detrended turnover

Group trading frictions **Source** Garfinkel (2009) **Frequency** monthly (daily inputs) **Agreement** 0.363

Inputs. dlyvol, shrout, primaryexch

Formula.

daily turnover, with Nasdaq volume scaled by 0.50 before 1997 and 0.38 after, less the cross-sectional mean that day, then detrended by subtracting its own median over the previous 180 trading days.

Notes. UNRESOLVED at 0.36, and the cause is known: the 180-trading-day median is not a moment of any month, so it cannot come out of the server-side aggregation the other eight daily characteristics use. It is currently approximated by a rolling median of nine monthly means. Full daily data has been pulled to compute the true median.

Rejected alternatives.

Variant	Agreement	Verdict
rolling median over 6, 9 and 12 months of monthly means	0.367	best available approximation so far

aBEME — Industry-adjusted book to market

Group value **Source** Asness, Porter and Stevens (2000) **Frequency** monthly **Agreement** 0.510

Inputs. be, cap, siccd

Formula.

$\text{aBEME} = \text{BEME} - \text{mean}(\text{BEME})$ within the firm's Fama-French 48 industry that month, equal-weighted over ordinary common shares that have BEME.

Notes. UNRESOLVED at 0.51, and the puzzle is that its parent BEME verifies at 0.97, so the adjustment itself is what fails. Our decile one matches (3.93 against a published 3.77) while decile ten does not (13.13 against 18.30): the published adjustment removes far less of the growth tilt than ours does. The search below has been exhaustive and nothing closes it. Most likely the industry classification comes from a source we do not have – plausibly Baba-Yara, Boons and Tamoni’s own file, since the paper says the characteristic set comes from that project as well as from FNW.

Rejected alternatives.

Variant	Agreement	Verdict
Fama-French 5, 10, 12, 17, 30, 38, 48 and 49 industries	0.521	ff30 marginally best, none close
one-digit, two-digit and three-digit SIC	0.477	rejected
Compustat historical SIC (sic), with and without CRSP fall-back	0.506	rejected
Compustat header SIC from comp.company	0.468	rejected
a firm’s modal SIC over its life	0.509	rejected
industry median instead of mean	0.511	rejected
capitalisation-weighted industry mean	0.227	rejected
industry mean over all of CRSP rather than common shares only	0.099	rejected
industry mean over NYSE firms only	0.384	rejected
industry mean over the larger half of each industry	0.401	rejected
ratio to the industry mean, and log of that ratio	0.440	rejected
standardised within industry (divided by the industry sd)	0.348	rejected
percentile rank within industry	0.421	rejected
cross-sectional rank less the industry mean rank	0.391	rejected
leave-one-out industry mean	0.512	rejected
aggregate industry ratio (sum of book over sum of market)	0.226	rejected
partial adjustment, subtracting 0.25, 0.5 or 0.75 of the industry mean	0.458	rejected
previous year’s industry mean	0.343	rejected

aSAT — Industry-adjusted asset turnover

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.447

Inputs. sale, at, siccd

Formula.

aSAT = SAT - mean(SAT) within the Fama-French 48 industry that month, equal-weighted over ordinary common shares.

Notes. UNRESOLVED at 0.45. Parent SAT verifies at 0.90. Same search as aBEME.

Rejected alternatives.

Variant	Agreement	Verdict
the full classification and operation search listed under aBEME	0.448	ff48 demeaning best, none close
aggregate industry ratio (sum of sales over sum of assets)	0.264	rejected
partial adjustment at 0.25, 0.5, 0.75	0.261	rejected

aPM — Industry-adjusted profit margin

Group profitability **Source** Soliman (2008) **Frequency** annual **Agreement** 0.284

Inputs. oiadp, sale, siccd

Formula.

aPM = PM - mean(PM) within the Fama-French 48 industry that month,
equal-weighted over ordinary common shares.

Notes. UNRESOLVED at 0.28, the worst of the industry-adjusted three even though parent PM verifies at 0.97. Separately, its one-month alpha is 0.00012 a month, statistically indistinguishable from zero, so which leg is "long" is not well defined for this characteristic in either dataset. Coverage is also poor in recent years (76% missing in 2024-25) and should be looked at.

Rejected alternatives.

Variant	Agreement	Verdict
the full classification and operation search listed under aBEME	0.340	two-digit SIC marginally best, none close
partial adjustment at 0.25, 0.5, 0.75	0.261	rejected
industry mean over the larger half, and the prior year's mean	0.146	rejected